InvariantCell: time-aware oncology evidence review

Phase 1 concept and feasibility candidate, v14. Internal draft, 27 September 2026. Applicant/team identity, rights and resource documentation still require review. No submission or registration has occurred.

An oncology research reviewer needs to distinguish an observation from a prediction, a proposed correction from an accepted revision, and a dated source from evidence actually available at a historical cutoff. A static graph can preserve each statement while obscuring these differences.

InvariantCell represents assertions with explicit provenance, polarity, context and separate time axes. A deterministic temporal engine returns supported answers or specific abstentions. An independent replay checks a certificate against the exact snapshot and query. These checks establish reproducibility within declared assumptions; they do not establish biological truth.

The core empirical result is temporal reasoning on a frozen final documentary panel: static lookup scored 24/138, timestamp filtering 42/138, and both temporal rules and verifier 138/138. The verifier contribution is inspectability, provenance, independent replay and typed failure; no incremental answer-accuracy gain over sound rules is claimed.

Evidence available includes a four-story, eleven-query documentary comparison; actual RDF/SHACL/bounded OWL-RL checks; and one independent image/reference measurement example. A new CIViC corpus contains 32 real revisions and 238 correlated queries, frozen before evaluation. Both strong rules and the verifier scored 238/238; all certificates replayed.

This document addresses temporal modeling (25%), reasoning correctness (25%), interoperability (20%), feasibility (15%) and workflow relevance (15%). References are included within ten pages. It is not submission-ready while expert review and component-level eligibility remain incomplete. [1]

1. A focused research evidence-review workflow

The intended user is a biomedical researcher reviewing molecular evidence and pathology measurements for an oncology question. Inputs are a molecular finding, stated disease/context, query time, permitted sources and the particular evidence question. Current demonstrations use public non-PHI material; no patient-specific treatment recommendation is generated.

Outputs are an attributed answer, supporting or opposing statements, relevant revision relationships, missing evidence and a replayable certificate. An abstention names the unmet condition, such as unknown historical availability, unknown specimen time, incomplete revision history or missing physical scale.

The workflow separates two questions. Retrospective review asks what a recorded revision says happened in the past, using today's source snapshot. Historical-knowledge reconstruction asks what evidence was publicly available then. A current snapshot containing old event timestamps can support the first question without establishing the second.

The new CIViC task asks whether a specific revision had editorial acceptance at a cutoff and which stored replacement value that revision accepted. The actual prior and suggested values are retained. It does not assume that a replacement remained the latest clinical interpretation or that the parent evidence item remained accepted.

A completed matched-access documentary M2 selected two of seven fixed predictions. Native, calibrated, rules and verified policies each selected one correct answer; rules and verifier selected the same pair. Verification did not improve this endpoint. The next scientific question needs real hidden measurements and outcomes, not a third repetition of this ranking comparison.

Current limits are substantive: no expert-adjudicated clinical decision set, no validated intervention sequence, no independent patient outcome and no demonstrated improvement over strong temporal rules. These are development requirements, not completed capabilities. [2,3]

2. Temporal representation and unknown information

An assertion includes a stable identity, subject, predicate, value, context, polarity, state kind and source digest. Observation and inference remain different state kinds. A source statement is retained as attributed evidence rather than silently promoted to biological fact.

Independent axes record event time, explicit validity interval, public evidence availability, local ingestion and prediction generation. Ontology extensions also retain publication and annotation/version times. Missing values remain unknown. Annotation, ingestion or database update dates cannot fill an unknown biological event or first-publication date.

Validity uses an explicitly declared half-open interval [start, end). A missing endpoint means an open bound only when an interval is explicitly asserted; otherwise validity is unknown. Nonpositive intervals are rejected. Date-only bounds in older documentary cases were conservative computational projections, not newly observed timestamps.

For CIViC, creation and acceptance timestamps describe curation events. Their intervals describe editorial acceptance of a revision. Biological time remains unknown. A conservative observed-public-by bound is recorded after receipt of the current source bodies. Unknown first-publication times prevent claims of exact historical public availability.

Corrections preserve predecessor and successor records. A negative qualifier remains attached to its assertion and is never flattened into a positive edge. Pending revisions are not automatically scientific conflicts. Conflicting support and refutation are reported only within the same stated proposition and scope.

This typed representation is implemented and reusable. It has not established a learned temporal embedding advantage, complete causal-path reasoning or arbitrary longitudinal trajectory reconstruction. Extension requires source-specific semantic validation and independently justified timing. [2,4]

3. Correctness, matched methods and certificates

A query specifies subject, predicate, context, public-evidence cutoff, ingestion cutoff and valid-at time, with an optional event cutoff. The engine excludes mismatched scope, unavailable evidence, invalid intervals, disallowed predictions and missing or late dependencies. It returns explicit support, refutation, conflict or abstention.

A certificate binds the snapshot, dependencies, query, result and algorithm version by digest. A separate implementation replays the claimed checks against the supplied snapshot. Tampering or inconsistent replay fails. A valid certificate can certify a justified abstention; it does not certify source completeness or biological correctness.

The fixed comparison uses identical typed inputs: static KG lookup; availability/ingestion timestamp filtering; strong rules adding interval and event constraints; and the existing verified temporal engine. No retrieval, source-access or fitting advantage is assigned to the verifier. The rule baseline is retained even when results tie.

New source gold is stored outside the solver interface. All predictions are materialized and hashed before the evaluator opens semantic gold. A once-only marker is acquired before comparator execution; a failed attempt remains consumed and its receipt persists. Separate development, validation and final-internal-test files make exposure explicit.

Metrics include exact answer correctness, unsupported availability uses, invalid-interval uses, curation-order violations, required-support coverage, provenance, abstentions, certificate replay and runtime. With unknown first-publication time, actual future-publication leakage is not identifiable. Missing multi-hop or biological-conflict tasks produce null metrics, not zero failures.

The v14 saved-proof audit replayed 138/138 final certificates with source-bound provenance on 138/138. All 552/552 specified copy-only attacks were rejected; 138/138 repeated replays agreed. All 69/69 abstentions were correct with typed reasons. Observed-cutoff, interval and curation-event violations were zero. With zero real contradiction witnesses, detection accuracy is not estimable. Historical first-publication leakage remains unknown. No answers were regenerated.

4. Empirical evidence and prospective evaluation

The completed v10 documentary comparison has four source-adjudicated stories and eleven queries. Static lookup answered 6/11 correctly, timestamp filtering 7/11, strong temporal rules 11/11 and verified temporal 11/11. Eleven certificates replayed. No accuracy improvement over strong rules was observed. These are related documentary queries, not independent clinical cases.

Older twenty-nine-query results remain archived and unchanged. No completed benchmark was rerun for this candidate. The v10 labels were written from primary sources independently of solver outputs by a coding agent, not validated by clinical experts. First-publication times were unknown. [3]

The new v13 source is actual CIViC revision history: 32 revisions, 87 attributed assertions and 238 queries. Development has 7 revisions/56 queries; validation 7/44; final internal test 18/138. Final candidates EID21-40 were specified before acquisition. Initial molecular families, profile IDs and publication IDs were excluded from final selection.

Stored prior and suggested field values, curation timestamps and exact source pointers make each case auditable. One initial therapy history is discontinuous, so general current-field reconstruction is not claimed. Current snapshots are not labeled historical archives. Code and selection were frozen before final intake; final answers are separate from policy inputs.

V13 exact correctness (development / validation / final): static 7/56, 10/44, 24/138; timestamp filtering 14/56, 17/44, 42/138; rules and verifier both 56/56, 44/44, 138/138. Both correctly abstained on 119 queries, with zero measured interval, availability or curation-event violations. All 238 certificates replayed. No incremental verifier accuracy gain was observed.

Rules took 1.029 seconds; verifier 5.521 seconds. Total run: 9.906 seconds, 71,460 KiB peak RSS. Final labels are now consumed. Queries remain correlated within 32 revisions; source-field adjudication is not clinical-expert gold. Historical public availability is unknown. Documentary correctness does not establish biological utility.

5. Ontology rigor and interoperability

Machine-readable artifacts include an RDF/XML ontology, SHACL Core shapes and a pinned Biolink mapping. They are bounded original projections, not complete imports of GO or Biolink. Non-hierarchical axioms cover source inheritance, inverse revision relations, symmetric contradiction, value restrictions and disjoint observed/predicted or positive/negative types. [4,5]

Executed v11 validation parsed the ontology and shapes, round-tripped 112 asserted data triples and checked graph isomorphism. Five attributed assertions conformed to actual SHACL checks. A 169-triple ontology with data yielded an 887-triple bounded OWL-RL closure, with no implemented-rule contradiction detected. The run took 4.712 seconds and 49,668 KiB maximum RSS.

Shape conformance and logical consistency are different checks. Missing required context can fail a shape without entailing a contradiction. Explicit disjoint types trigger the implemented logical diagnostic. Full OWL completeness, exhaustive unsatisfiable-class analysis and all possible datatype/profile errors remain unevaluated.

GO/GAF handling preserves NOT, other qualifiers, evidence codes, annotation dates, namespace, obsolete status and replacement metadata. Biolink identifiers are checked against version 4.3.6. There is no matching Sample class in that pin; local Specimen remains unmapped. Broad relationship projections retain exact local predicates and do not imply complete Biolink instance conformance.

Internal KG formulas are versioned provisional_internal_v1, not organizer scores. Recorded values are: 112 data triples; relationship richness 0.611111; attribute richness 1.0; class instantiation 3/23; property instantiation 14/36; internal interlinking 5/7. Full unsatisfiable-class count is unknown, not zero. [4]

The new revision profile passed typed/source integrity checks. It has not undergone a new profile-specific OWL/SHACL study; earlier formal results are not automatically transferred to every later corpus.

6. A real quantity-specific pathology connection

One public NuCLS breast-cancer field provides an independent image/reference demonstration. The original RGB image and corrected reference annotations come from the source authors. The fixed image-only baseline writes predictions before the scorer reads reference arrays. No parameter was retuned after evaluation. This asset is distinct from P09 and has no asserted molecular pairing. [6]

Actual results on eligible references: 46 predicted objects versus 42 reference nuclei; 19 matched objects, 27 false positives and 23 missed objects; object F1 0.431818. Pixel Dice is 0.780564 and IoU 0.640102. Near count agreement therefore hides poor localization.

Eligible predicted area is 18,516 pixels versus 19,928 reference pixels. At the source-declared rendered scale of 0.2 micrometers per pixel in both axes, this is 740.64 versus 797.12 square micrometers. These areas cover eligible pixel-reference regions, not all nuclei, whole-slide tissue or patient-level area.

Count/localization references and pixel/area references differ. Rectangle-only and unknown regions are excluded from pixel scoring; valid rectangles can contribute to object matching. Count is not inferred from area. Physical units require a declared scale and source; absent scale stays in pixels.

The typed measurement bridge retains reference count 42 as observed and baseline count 46 as inferred. Specimen/event and source-availability times remain unknown, so the temporal replay conservatively abstains where required. A cross-sectional image is not manufactured into a longitudinal case.

A later six-field, three-participant trial failed its completion check. Separate posthoc reference recovery measured two fields from two participants: object F1 0.2921/0.2951 and count error 39 in each. Four dimension failures remain; the full patient macro is undefined. This is not a rescued trial. A new patient-disjoint benchmark and independent quantity-specific references remain necessary.

7. Implementation feasibility and Phase 2 package

The Python implementation separates source ingestion, typed evidence, temporal evaluation, independent replay and optional ontology checks. Plain JSON records and hashes permit inspection without a model service. Frozen inputs, outputs and receipts support offline replay and exact provenance review.

The CIViC acquisition used 1,742,054 payload bytes. New ClinVar intake added 771,328 bytes: seven exact accession-version responses, two silently mismatched responses rejected, and two three-version series. Classification values remain uninspected, gold pending. Current retrievals of archived record versions are not byte-identical historical release snapshots; publication times stay unknown. [7]

The completed comparison used one CPU with a two-GiB, 180-second bound, without network or fitting. Actual run time was 9.906 seconds with 71,460 KiB peak RSS. Its once-only attempt is consumed. Predictions were saved before gold evaluation; the prediction digest was independently verified. New source review must not be presented as fresh confirmation of this consumed panel.

Optional formal checks used project-local RDFLib 7.6.0, pySHACL 0.40.1 and OWLRL 7.6.2. The base environment was not modified. Their licenses, versions and contribution/funding provenance belong in the final dossier; a successful import is not eligibility clearance.

Phase 2 preparation includes executable code, split data, immutable snapshots, configurations, dependency pins and documented hardware bounds. An independent expert review, wider natural-history corpus, complete profile-specific interoperability checks and a recorded walkthrough remain pending. No NIH-supplied data is assumed. [1]

Extensibility is tested by preserving semantics across documentary evidence and pathology quantities, with source-specific adapters. Further molecular/assay cases must supply actual identity, temporal scope and independent outcomes; interface reuse alone is not biological transfer evidence.

8. Assumptions, provenance and readiness gates

Core assumptions are explicit: source artifacts are correctly identified; current revision API fields faithfully preserve the recorded curation events; asserted intervals describe editorial states only; and every negative or inferred statement retains its scope. These assumptions are auditable but not proven by certificates.

Known gaps include incomplete historical availability, discontinuous field histories, no clinical-expert gold, sparse independent families, no real unresolved biological-conflict benchmark, no intervention sequence and no independent patient endpoint. Formal gaps include full OWL consistency and an exhaustive unsatisfiable-class census. Expected ties and negative findings remain reportable outcomes.

CIViC content is CC0 and inspected source code MIT. NuCLS data is CC0 with separately licensed code. Original local code, retained upstream artifacts and optional libraries require a component-level rights/provenance inventory. Data reuse permission and non-PHI status do not by themselves establish challenge eligibility.

The user attests nonfederal resources; documentary compute, development/support, contributor and institutional evidence remains pending. Installed dependency licenses and data reuse terms are inventoried separately, including NCBI submitter-IP caveats and the local PDF renderer license. Public access does not prove prize eligibility. Earlier work is not retrospectively cleared. [1]

The Phase 1 window is October 5, 2026 to January 15, 2027. Phase 2 is April 5 to November 12, 2027, and limited to Phase 1 winners. Registration/terms acceptance and submission require separate user action; neither has been performed here. No exact closing hour beyond the published time-zone statement is invented. [1]

Before a submission-ready label: complete source/expert review, assess the completed rule tie and verification cost, reconcile current requirements, finish the eligibility and rights dossier, confirm authorship and verify the final PDF. This candidate records a feasible bounded prototype and an honest evaluation, not a clinical or prize claim.

References and reproducible artifact index

[1] NIH official challenge announcement, verified 2026-09-27. https://www.nih.gov/challenges/its-about-time-temporal-reasoning-biomedical-knowledge-graphs-challenge

[2] CIViC primary API and curation documentation. https://docs.civicdb.org/en/latest/api.html ; https://docs.civicdb.org/en/latest/curating/evidence.html

[3] Archived v10 evidence: reports/nih/independent_case_protocol.md ; runs/post_confirmation_v10/dated-cases-v1/results.json. No rerun.

[4] InvariantCell v11 formal artifacts and actual receipt: ontology/nih_v11.owl ; ontology/nih_v11_shapes.ttl ; ontology/biolink_mapping_v11.yaml ; reports/nih/ontology_v11_validation.json. Formulas: provisional_internal_v1.

[5] W3C SHACL and OWL 2 profiles: https://www.w3.org/TR/shacl/ ; https://www.w3.org/TR/owl2-profiles/ . Biolink pinned source: https://github.com/biolink/biolink-model/tree/v4.3.6

[6] NuCLS primary data, format and license: https://sites.google.com/view/nucls/single-rater ; https://sites.google.com/view/nucls/data-format ; https://github.com/PathologyDataScience/NuCLS . Actual report: reports/pathology/real_measurement_validation.md

CIViC semantic source and CC0/MIT statement pinned at https://github.com/griffithlab/civic-v2/tree/0488d61aeb0a5c45d2a1a7edc114ac085942939f . Sources were inspected, not executed. Current code does not certify lossless historical migration.

V13 freeze and results: data/post_confirmation_v13/nih_revision/freeze.json ; runs/post_confirmation_v13/nih-civic-revisions-v1/results.json. V14 proof audit and provenance inventory: reports/nih/v14/ . No original answers were regenerated.

[7] ClinVar version retrieval and identifier semantics: https://www.ncbi.nlm.nih.gov/clinvar/docs/maintenance_use/ ; https://www.ncbi.nlm.nih.gov/clinvar/docs/identifiers/ . Reuse policy: https://www.ncbi.nlm.nih.gov/home/about/policies/ . Gold pending.